DA1300+DA1301 — Programming and Data Structures (+ Python lab)

Tier	A
Phase (order)	1
Do this first	Nothing
Estimated effort	10-12 weeks, ~8 h/week
Unlocks	DA2303, DA2304, DA2400, everything

Roadmap

1. Python basics: types, lists, dicts, functions, classes.
2. Linked lists, queues, trees, sorting/searching, priority queues, implemented as classes.
3. NumPy and Matplotlib for arrays and plots.

Video courses (all links checked on YouTube on 2026-10-22)

🎓 Main course: follow it start to end

- Programming in Python (95 videos). IIT Madras BS *Programming in Python* (Sudarshan Iyengar): IITM's own Python course.

🎥 Other full lecture series (pick one if the main one doesn't click, or for topics it skips)

- Programming, Data Structures And Algorithms Using Python (45 videos). NPTEL *Programming, Data Structures and Algorithms using Python* (Madhavan Mukund): the data-structures half.
- CS50's Introduction to Programming with Python (CS50P) 2022 (11 videos). Harvard CS50P (2022, official): polished, with problem sets you can autograde.

⚡ Short videos for intuition and revision

- Python Tutorials (100+ videos). Corey Schafer *Python Tutorials*: look up any single topic (decorators, generators, OOP, venvs).

Study plan (week by week)

Lecture numbers are positions in the main playlist above. Click a number to open that video.

Week	Lectures	What to do
1	1–16	Variables, literals, data types, strings and slicing, operators, input. Install Python locally as well as using Replit.

Week	Lectures	What to do
2	17–25	String methods, escape characters, if/elif/else, importing libraries.
3	26–38	while and for loops, range, nested loops, formatted printing. Solve 15 loop problems (e.g. first page of CodingBat Python).
4	39–52	Problem solving with lists: birthday paradox, dot product, matrix add/multiply, naive sorting and search, first functions.
5	53–60	Lists, tuples, dictionaries, sets and when to use each. Write a word-frequency counter for a book from Project Gutenberg.
6	61–65, 68–69	Function arguments, scope, iterators/generators, lambda, map/filter, enumerate/zip.
7	66–67, 70–74	File handling, big files, a Caesar cipher on a file. Parse a CSV by hand, then with pandas.
8	75–86	Recursion, recursive sorting, binary search (iterative and recursive), exceptions.
9	84, 87, 90–91	OOP: classes, objects, attributes, methods, inheritance. Model a library system with classes.
10	88–89, 92–95	NumPy, pandas, matplotlib. Mini-project: load a real dataset (e.g. from Kaggle), clean it and plot 3 charts.

▼ **All lectures in *Programming in Python* (95 videos)**

1. L1: A quick introduction to variables | programming in python
2. L2: Introduction to programming basics in python
3. L3: Introduction to REPLIT | programming in python
4. L4: More on REPLIT, print command & common python mistakes
5. L5: Variables & literals in python
6. L6: Data types part 1 | programming in python
7. L7: Data types part 2 | programming in python
8. L8: Introduction to strings & string slicing in python
9. L9: Operators & expressions | part 1 | programming in python
10. L10: Operators & expressions | part 2 | programming in python
11. L11: More on strings | programming in python
12. L12: Variables - a programmer's perspective | programming in python
13. L13: Variables & input statement | programming in python
14. L15: Variables revisited dynamic typing | programming in python
15. L16: Introduction | programming in python
16. More on Variables, Operators and Expressions
17. L17: Mastering string methods in python
18. L18: Escape characters & types of quotes in python strings
19. L19: Conclusion | FAQs & tips for beginner python programmers

20. L20: An interesting cipher | more on strings | programming in python
21. L21: Introduction to the if statement | programming in python
22. Tutorial on if, else and else-if (elif) conditions
23. L23: Different ways to import a library | programming in python
24. L24: Introduction | programming in python
25. L25: Conclusion python syntax variables if statements & libraries
26. L26: Introduction to while loop | programming in python
27. L27: Introduction to import library | programming in python
28. while to Compute Factorial
29. L28: FOR loop to calculate sum of first n numbers in python
30. L30: Introduction to for loop in python with examples
31. Tutorial on while loop | programming in python
32. L32: For loop for multiplication tables | programming in python
33. L33: More on range & for loop without range | programming in python
34. L34: Nested for loops & string operations in python
35. Tutorial on for loop & difference between while loop & for loop
36. L36: Conclusion | programming in python
37. Tutorial on nested loops | programming in python
38. L38: Formatted printing
39. L39: Introduction to problem solving practice in python
40. L40: Warmup with lists | programming in python
41. L41: Break continue & pass in python explained with examples
42. L42: Birthday paradox in python with random module
43. L43: Dot product of vectors in python using for loops
44. L44: The obvious sorting in python
45. L45: Matrix addition in python from scratch
46. L46: Matrix multiplication | part 1 in python step by step
47. L47: Naive search in a list | programming in python
48. L48: Matrix multiplication | part 2 | programming in python
49. L49: More examples of functions | programming in pPython
50. L50: Sorting in python using functions refactoring the obvious sort
51. L51: Introduction to functions | hands-on coding with examples
52. L52: Conclusion | mastering fundamentals & preparing for challenges
53. L53: Lists & sets | understanding trade-offs & efficiency
54. L54: Tuples | immutable data structures explained
55. L55: Dictionaries | efficient data storage & retrieval
56. L56: Matrix multiplication using functions | modular & clean approach
57. L57: More on lists | advanced operations & methods
58. L58: More on tuples | immutability, packing & hashable tuples
59. L59: More on dictionaries | keys, values & essential methods
60. L60: More on sets | operations, properties & methods

61. L61: Types of function arguments
62. L62: Functional programming | part 2 | inline statements & list comprehensions
63. L63: Types of functions | built-in, library, string & user-defined
64. L64: Scope of variables
65. Tutorial on functions
66. L66: Introduction | advanced concepts & exception handling
67. L67: Reading & writing to a file | basics of file handling
68. L68: Functional programming | part 1 | iterators, generators, & yield
69. L69: Functional programming | part 3 | lambda, enumerate, zip, map, filter
70. L70: Very big files a tip
71. L71: Caesar cipher | file encryption example with sherlock holmes
72. L72: Big text file handling | searching & error management
73. L73: Introduction to file handling | basics & pandas overview
74. L74: File handling, genetic sequences | f.seek, f.read & knuth morris pratt algorithm
75. L75: Introduction to the week & recursion | concepts & applications
76. L76: Theoretical introduction to recursion
77. L77: Recursion - an illustration | sum of n numbers, factorial & more
78. L78: Recursion: a simple question | checking lLists with recursive logic
79. L79: Recursion: find 0 in a list | practical example & vim/ipython setup
80. L80: Sorting recursively | recursive sorting algorithm explained
81. L81: Binary search implementation | efficient searching explained
82. L82: Exception handling | try, except, finally & custom errors
83. L83: Binary search recursion way | recursive algorithm explained
84. L84: Introduction to OOP | object-oriented programming basics
85. L85: Warm up for binary search | obvious search & time complexity
86. L86: Introduction to binary search | halving technique & real-world examples
87. L87: Inheritance & method overriding | types of inheritance & file importing
88. L88: Introduction to numpy library | arrays vs lists explained
89. L89: Why pandas | data analysis made easy
90. L90: Attributes & methods | OOP, init, self, and behavior explained
91. L91: Classes & objects | object oriented programming fundamentals explained
92. L92: Pandas series, dataframe & more | data analysis made easy
93. L93: Course summary | review of core python concepts & next steps
94. L94: Introduction to matplotlib library | data visualization with charts & plots
95. L95: How to submit a programming assignment

Books, notes and websites

- ✅ NPTEL *Programming, Data Structures and Algorithms using Python* (Madhavan Mukund, CMI) — elearn.nptel.ac.in/?p=422
- ✅ NPTEL *Programming and Data Structures* (N.S. Narayanaswamy, IIT Madras) — nptel.ac.in/courses/106106130/

- Book: Goodrich, Tamassia, Goldwasser, *Data Structures and Algorithms in Python* (the lab text)
- CS231n Python/NumPy tutorial; Matplotlib docs
- Practice: LeetCode, 3 problems/week

Practice / proof of understanding

Implement linked list, queue, BST, heap and a sort in Python, plus a NumPy script that plots a function and its gradient.

What your IITM course sheet says

From the PDF: DA1300

Official description: The course intends to cover a solid foundation of Programming and Data Structures in Python. The learning outcomes of the course include developing a basic understanding of Python, ability to understand various data structures in Python, understand and analyze basic algorithms (sorting/searching).

Official syllabus (trimmed):

> • Python Primer • Object Oriented Programming • Array based Squences • Stacks, Queues, Deques • Linked List, Trees, Priority Queues • Sorting/Searching

Official textbooks: Data Structures and Algorithms in Python by Goodrich, Tamassia and Goldwasser, Wiley.

From the PDF: DA1301

Official description: The course intends to serve as a laboratory companion course to the Programming and Data Structures course. The learning outcomes of the course include developing a basic understanding of Python, ability to implement basic data structures and basic algorithms (sorting/searching) in python and a good understanding of using Numpy and matplotlib.

Official syllabus (trimmed):

> • Introduction to Python basics, keywords • Basic python data types, list, tuple, dictionary, indexing • Object oriented implementation of linked lists, queues, trees • Sorting and Searching • Priority queues • Numpy and matplotlib

Official textbooks: Data Structures and Algorithms in Python by Goodrich, Tamassia and Goldwasser, Wiley.

From the PDF: DA5300

Official description: This course aims to teach different topics in data structures and algorithms for data science in Python.

Official syllabus (trimmed):

1. ADT Array, Linked Lists, Stacks, Queues
2. ADT Binary Trees
3. ADT Dictionary: Binary search trees, balanced binary search trees - AVL Trees. Hashing collisions, open and closed hashing
4. ADT Priority queues: Binary heaps
5. Graphs: Representations (Matrix and Adjacency List), basic traversal techniques: Depth First Search + Breadth First Search (Stacks and Queues)
6. Tensors and their applications in Neural networks.
7. Introduction to approximate similarity (neighbor) search data structures - multidimensional index structures (kd-trees, R trees, quad trees)
8. Introduction to Probabilistic Data Structures - bloom filter, count-min, hyperloglog, locality-sensitive hashing.

Official textbooks: 1. Data Structures and Algorithms in Python by Michael H. Goldwasser, Michael T. Goodrich, and Roberto Tamassia, Wile 2013.

Official references: 1. Horowitz Ellis, Sartaj Sahni, Sanguthevar Rajasekaran (2007) Computer Algorithms, 2/e, Silicon Press 2.Thomas H. Cormen, Charles R. Leiserson CR, Ronald L. Rivest, Clifford Stein (2009), Introduction to Algorithms, 3/e, Prentice Hall India 3. Skiena SS (2010) The Algorithm Design Manual, Springer, ISBN 1849967202